Statistical Analysis of E-Commerce Transaction Data

Mobin Abedian
Student ID: 403222152
mobinabedian6@gmail.com

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Abstract

This report presents a statistical analysis of an e-commerce transaction dataset containing over 530,000 records. A series of hypothesis tests are conducted to examine revenue distributions, pricing patterns across countries, and associations between transaction characteristics. Key findings reveal that revenue is not normally distributed, there is a moderate negative correlation between quantity and unit price, and significant differences exist in revenue across countries.

Contents

1 Introduction

Understanding customer behavior and transaction patterns is fundamental for data-driven e-commerce strategy. This study applies classical statistical hypothesis testing to an online retail dataset to answer specific business questions about pricing, revenue, and cross-country purchasing behavior.

The analysis addresses the following questions:

- Does transaction revenue follow a normal distribution?
- Is there an association between quantity ordered and unit price?
- Do average prices differ between the UK and France?
- Is there a significant revenue difference between UK and international transactions?
- Do mean revenues differ across top countries?
- Is there an association between country and high-value purchases?

2 Dataset and Preprocessing

2.1 Data Source

The dataset contains transactional records from a UK-based online retailer between 2010 and 2011. It includes 541,909 records with variables such as InvoiceNo, StockCode, Description, Quantity, InvoiceDate, UnitPrice, CustomerID, and Country.

2.2 Data Cleaning

Records with non-positive Quantity or UnitPrice were removed, leaving 530,104 clean transactions. A derived `revenue` column was created as Quantity $\times$ UnitPrice. The UK dominates the dataset with 485,123 records, while all other countries combined account for 44,981 records.

3 Methodology

All statistical tests were conducted using Python with `scipy.stats` at a significance level of $\alpha = 0.05$:

- **Shapiro-Wilk test:** normality assessment
- **Spearman rank correlation:** monotonic association between variables
- **Levene's test:** variance homogeneity check
- **Independent t-test / Welch's t-test:** two-group mean comparison
- **One-way ANOVA:** multi-group mean comparison
- **Chi-square test of independence:** categorical association

4 Results

4.1 Test 1: Normality of Revenue

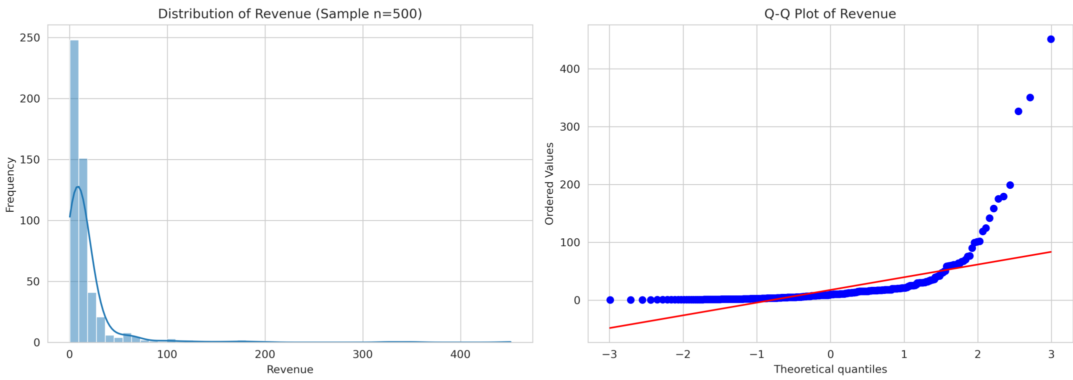

Figure 1: Histogram with KDE and Q-Q plot of revenue (sample n=500). The distribution deviates significantly from normality.

Shapiro-Wilk: $W = 0.4037$, $p \approx 0.0000$

Since $p < 0.05$, we reject H_0. Revenue is not normally distributed. This justifies the use of non-parametric methods in subsequent tests where appropriate.

4.2 Test 2: Quantity and Unit Price Association

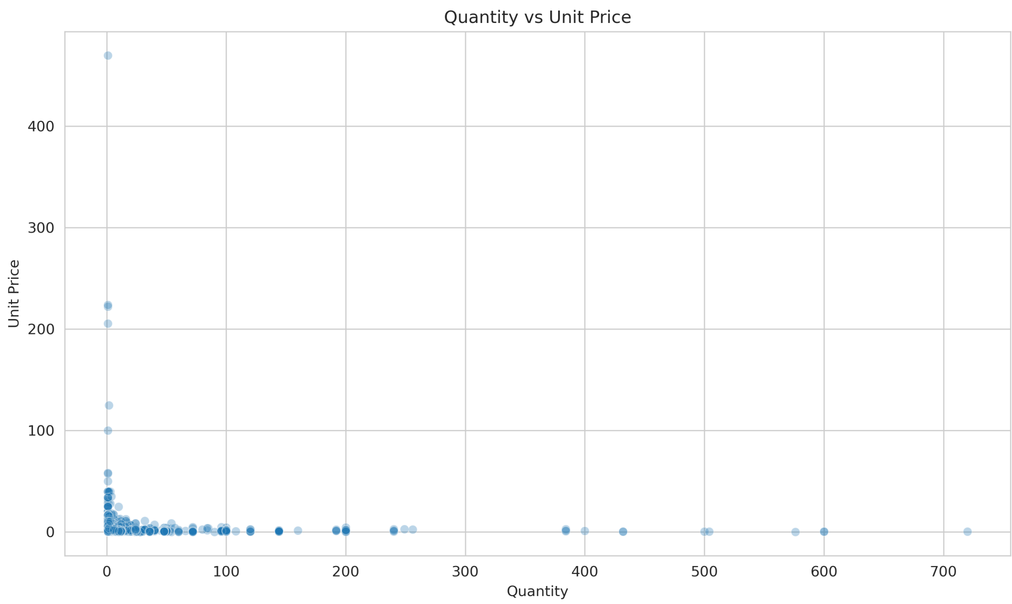

Figure 2: Scatter plot of Quantity vs Unit Price with 5,000 sampled records.

Spearman correlation: $\rho = -0.4032$, $p \approx 0.0000$

There is a statistically significant moderate negative monotonic association. As unit price increases, quantity ordered tends to decrease. This aligns with standard economic demand theory.

4.3 Test 3: Price Comparison – UK vs France

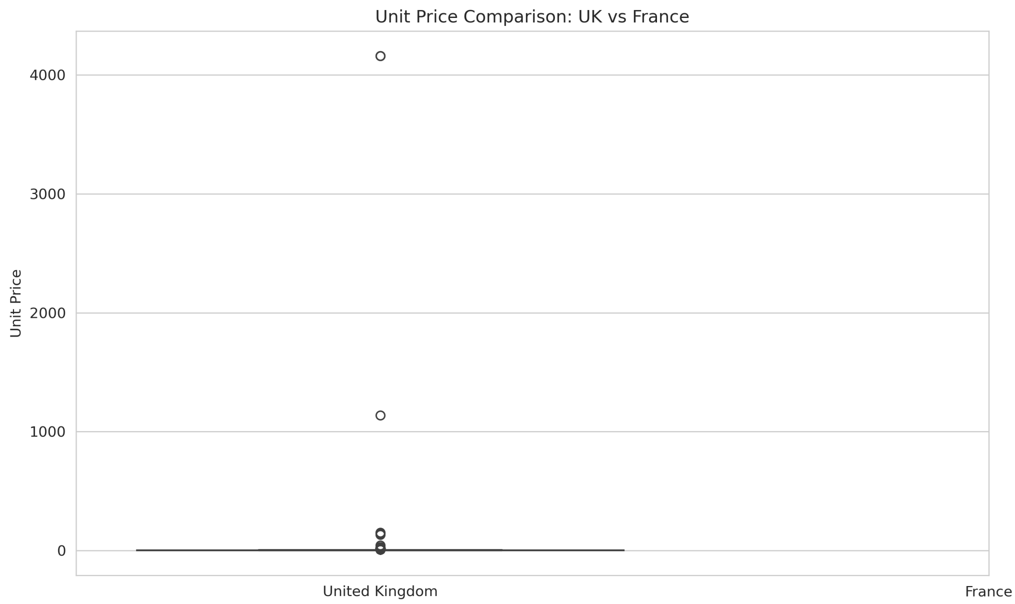

Figure 3: Box plot comparing unit prices between the UK and France.

Levene's test: $p = 0.0886$ (variances are equal)

Independent t-test: $t = -1.4127$, $p = 0.1577$

We fail to reject H_0. There is no statistically significant difference between average unit prices in the UK and France. The observed difference is attributable to random variation.

4.4 Test 4: Revenue – UK vs International

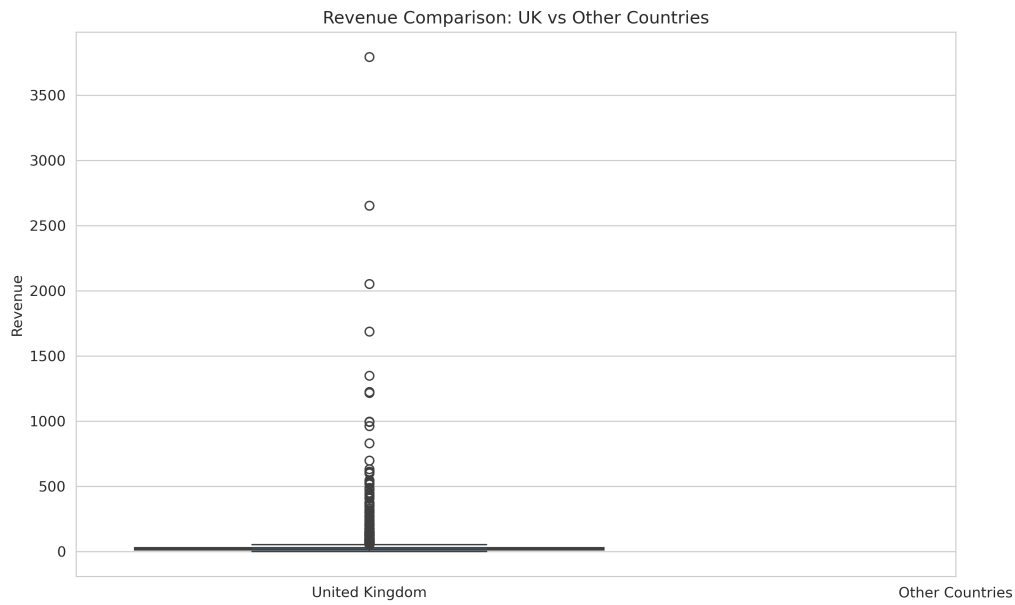

Figure 4: Box plot comparing revenue between UK and international transactions.

Levene's test: $p \approx 0.0000$ (unequal variances)

Welch's t-test: $t = -30.8258$, $p \approx 0.0000$

We reject H_0. International transactions generate significantly higher average revenue ($\bar{x} = 36.49$) compared to UK transactions ($\bar{x} = 18.60$). This is the largest effect observed in the analysis.

4.5 Test 5: Revenue Across Top 5 Countries

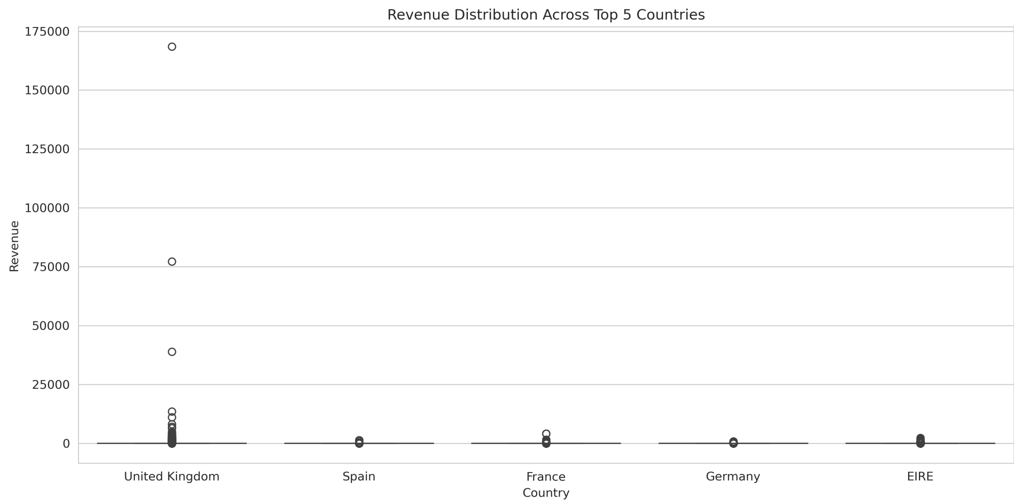

Figure 5: Revenue distribution across the top 5 countries by transaction volume.

One-way ANOVA: $F = 10.2228$, $p \approx 0.0000$

We reject H_0. At least one country has a significantly different mean revenue. EIRE (Ireland) shows notably higher average revenue ($\bar{x} = 35.93$) compared to the UK ($\bar{x} = 18.60$), while Germany, France, and Spain cluster around 24–25.

4.6 Test 6: Country and High-Value Purchases

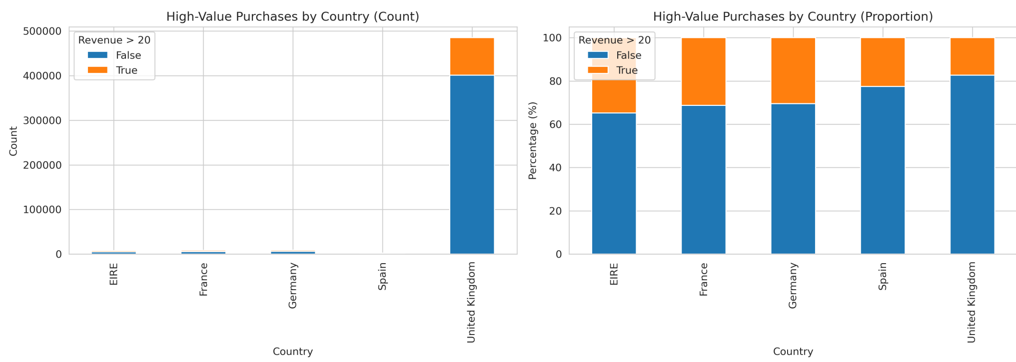

Figure 6: Count and proportion of high-value purchases (revenue > 20) by country.

Chi-square test: $\chi^2 = 3661.10$, $\text{df} = 4$, $p \approx 0.0000$

We reject H_0. There is a statistically significant association between country and the likelihood of a high-value purchase. International customers, particularly from EIRE, have a higher proportion of high-value orders compared to UK customers.

5 Discussion

5.1 Summary of Findings

Test	**Method**	**Result**
Revenue normality	Shapiro-Wilk	Not normal ($p < 0.001$)
Qty–Price association	Spearman	Moderate negative ($\rho = -0.40$, $p < 0.001$)
UK vs France price	t-test	No difference ($p = 0.158$)
UK vs Intl revenue	Welch's t-test	Significant ($p < 0.001$)
Revenue across countries	ANOVA	Significant ($p < 0.001$)
Country vs High-value	Chi-square	Significant ($p < 0.001$)

Table 1: Summary of all statistical test results

5.2 Interpretation

The most impactful finding is the substantial revenue gap between UK and international transactions. International customers spend nearly twice as much per transaction on average. This could reflect differences in shipping costs, basket composition, or customer segments. The lack of price difference between UK and France suggests that pricing strategy is consistent across these markets.

5.3 Limitations

- The dataset is heavily imbalanced toward UK transactions (91.5%)
- ANOVA does not identify which specific pairs differ (post-hoc tests were not conducted)
- The chi-square test is sensitive to the large sample size, which may produce statistically significant results for practically negligible effects
- No correction for multiple comparisons was applied

6 Conclusion

This statistical analysis of e-commerce transaction data reveals clear and significant patterns. International transactions consistently generate higher revenue than domestic UK transactions, and country of origin is strongly associated with purchasing behavior. The moderate negative correlation between quantity and price confirms standard economic intuition. These findings can inform targeted marketing strategies, pricing adjustments, and international expansion planning.

A Appendix

A.1 Code Repository

The complete analysis code is available in the Jupyter notebook `e_commerce_statistical_analysis.i`
All figures were generated using `generate_figures_ecommerce.py`.

The full project is hosted on GitHub: https://github.com/fermow/SBU-Data-Science-2026

A.2 Variable Descriptions

Variable	Description
InvoiceNo	Invoice identifier
StockCode	Product stock code
Description	Product description
Quantity	Number of units purchased
InvoiceDate	Date and time of transaction
UnitPrice	Price per unit
CustomerID	Customer identifier
Country	Customer country
revenue	Derived: Quantity $\times$ UnitPrice
is_high_value	Derived: revenue $>$ 20

Table 2: Variable descriptions

A.3 Technology Stack

- Python 3.13
- pandas, numpy
- SciPy (statistical tests)
- matplotlib, seaborn (visualization)
- LaTeX (pdfLaTeX)